



EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

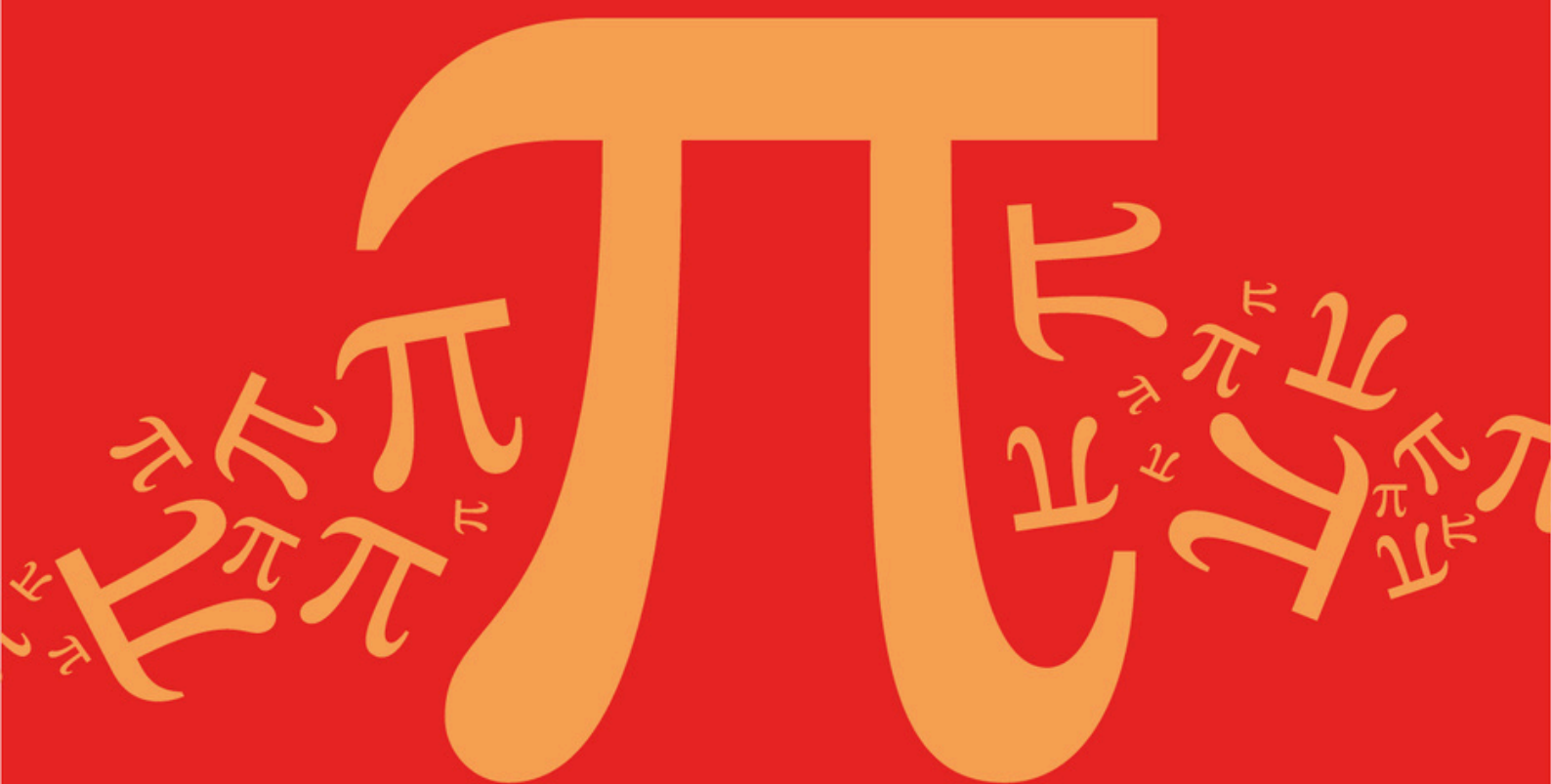
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

B: Complex Numbers

Sub topic

B6: Multiplication and Division in Modulus-Argument Form

Mark Scheme

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B6: Multiplication and Division in Modulus-Argument Form

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	$\frac{\pi}{12}$
Total 1 mark			

Q2.

(a) (i) $1 + \sqrt{3}i = 2e^{\frac{\pi i}{3}}$
B1 both correct

B1

$1 - i = \sqrt{2}e^{-\frac{\pi i}{4}}$
OE

B1B1

3

(ii) $2^{\frac{21}{2}}$ or equivalent single expression
No decimals; must include fractional powers

B1F

Raising and adding powers of e

M1

$\frac{17\pi}{12}$ or equivalent angle
Denominators of angles must be different

A1F

3

(b) $z = \sqrt[3]{2^{10}\sqrt{2}} e^{\frac{17\pi i}{36} + \frac{2k\pi i}{3}}$

M1

$\sqrt[3]{2^{10}\sqrt{2}} = 8\sqrt{2}$

CAO

B1

$$\theta = \frac{17\pi}{36}, -\frac{7\pi}{36}, -\frac{31\pi}{36}$$

Correct answers outside range: deduct 1 mark only

A2,1F

4

[10]

Q3.

(a) $\frac{1+i}{1-i} = \frac{(1+i)^2}{1-i^2} = i$
AG

M1A1

2

(b) $|z_2| = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1 = |z_1|$

M1A1

2

(c) $r = 1$
PI

B1

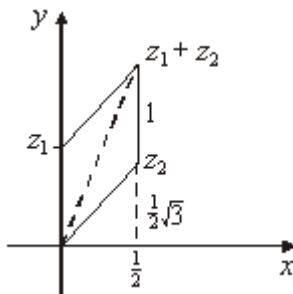
$$\theta = \frac{1}{2}\pi, \frac{1}{3}\pi$$

Deduct 1 mark if extra solutions

B1B1

3

(d)



Positions of the 3 points, relative to each other, must be approximately correct

B2,1F

2

(e) $\text{Arg}(z_1+z_2) = \frac{5}{12}\pi$

Clearly shown

B1

$$\tan \frac{5}{12} \pi = \frac{1 + \frac{1}{2}\sqrt{3}}{\frac{1}{2}}$$

Allow if B0 earned

M1

$$= 2 + \sqrt{3}$$

AG must earn B0 for this

B1

3

[12]



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Examiner reports

Q2.

Part (a)(i) was generally well done. The less successful candidates usually wrote the argument of $1 - i$ as $3\pi/4$ instead of $-\pi/4$. In part (a)(ii) there was some poor handling of fractions in the argument of the product of the two complex numbers, and also some omission of raising the moduli of the two complex numbers to their respective powers. Many of the candidates who had been successful in part (a) often went on to complete part (b) correctly, although some candidates lost marks either through not giving z in the form asked for or by giving values for θ outside the specified range.

Q3.

Part (a) was well done, as was part (b). In part (c) it was surprising to note how many candidates could not express the complex number in the form $re^{i\theta}$, although z_2 was almost invariably correctly written as $e^{i\frac{\pi}{3}}$. Errors in part (c) however did not deter candidates from drawing a correct Argand diagram as they usually used the form $a + ib$ when plotting their points. Although the vast majority of scripts ended with $\tan \frac{5\pi}{12} = 2 + \sqrt{3}$, very, very few candidates gave convincing proof that $\arg(z_1 + z_2)$ was $\frac{5\pi}{12}$, but rather seemed to take it for granted.